

A Three-Loop Magnetic-Field Array for Polarization-Independent HF Reception

Geometry, Direction-Dependent Projection, Polarimetry, and a Validation Pipeline

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Abstract

We describe a compact three-element magnetic-loop antenna array in which the loops sit on three faces of a cube whose body diagonal is vertical and whose lower vertex rests on the ground. This geometry makes the three loop normals identically tilted from the zenith and uniformly spaced in azimuth. We derive the linear transformation that takes the three measured magnetic-field components to the lab-frame magnetic field, and from there to the wave’s magnetic field projected onto a plane perpendicular to a chosen arrival direction. The resulting $|\mathbf{B}_\perp|^2$ is a polarization-independent intensity estimator; the same projection plus a complex-baseband extraction yields the Stokes parameters and hence the polarization state of the wave. A companion Python module implements the full pipeline and is validated on a simulated 20 MHz signal with controllable Faraday rotation and additive noise.

1 Coordinate systems and array geometry

We work in a right-handed lab frame with axes (E, N, Up) : East, North, and the local vertical pointing up. The arrival direction of a plane wave is parameterized by an azimuth φ , measured clockwise from North (so $\varphi = 0^\circ$ is N, $\varphi = 90^\circ$ is E), and an elevation θ measured upward from the local horizontal. The unit vector pointing *toward the source* is

$$\hat{\mathbf{k}}(\varphi, \theta) = (\cos \theta \sin \varphi, \cos \theta \cos \varphi, \sin \theta)^\top. \quad (1)$$

Figure 1 shows the convention.

Cube on a vertex

Consider a unit cube standing on its $(0, 0, 0)$ corner with its body diagonal aligned to the lab-frame vertical. The three faces meeting at that lower vertex have outward normals each tilted $\arccos(1/\sqrt{3}) = 54.7356^\circ$ from the zenith (equivalently, 35.26° above the horizontal), and their horizontal projections are spaced 120° apart in azimuth. The three loops are placed flat on those faces, so the i -th loop senses the magnetic-field component along the corresponding face normal $\hat{\mathbf{n}}_i$.

We label the loops L1, L2, L3 with horizontal-projection azimuths of 0° , $+120^\circ$, -120° respectively. Loop L1’s normal points “N + up”; L2’s points roughly “S-of-east + up”; L3’s points roughly “S-of-west + up”. In lab coordinates,

$$\mathbf{N} = \begin{pmatrix} \hat{\mathbf{n}}_1^\top \\ \hat{\mathbf{n}}_2^\top \\ \hat{\mathbf{n}}_3^\top \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{2/3} & \frac{1}{\sqrt{3}} \\ +\sin 54.74^\circ \sin 120^\circ & \sin 54.74^\circ \cos 120^\circ & \frac{1}{\sqrt{3}} \\ -\sin 54.74^\circ \sin 120^\circ & \sin 54.74^\circ \cos 120^\circ & \frac{1}{\sqrt{3}} \end{pmatrix}. \quad (2)$$

Azimuth / elevation convention
az measured CW from N; el above horizontal

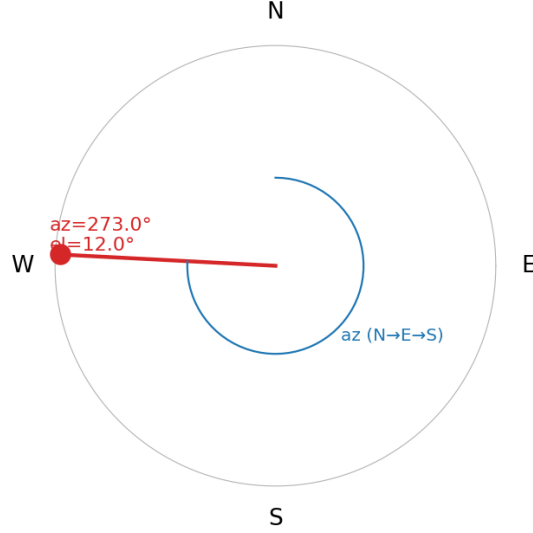


Figure 1: Azimuth/elevation convention. φ is measured clockwise from N (so the sequence is $N \rightarrow E \rightarrow S \rightarrow W$); θ is the elevation above the horizontal. The red dot illustrates a wave arriving from $\varphi = 273^\circ$, $\theta = 12^\circ$, the approximate path of WWV $\rightarrow$ Cambridge MA.

Figures 2a and 2b illustrate the geometry.

From three measurements to the lab-frame $\mathbf{B}$ vector

Each loop senses $B_i(t) = \hat{\mathbf{n}}_i \cdot \mathbf{B}_{\text{lab}}(t)$, so

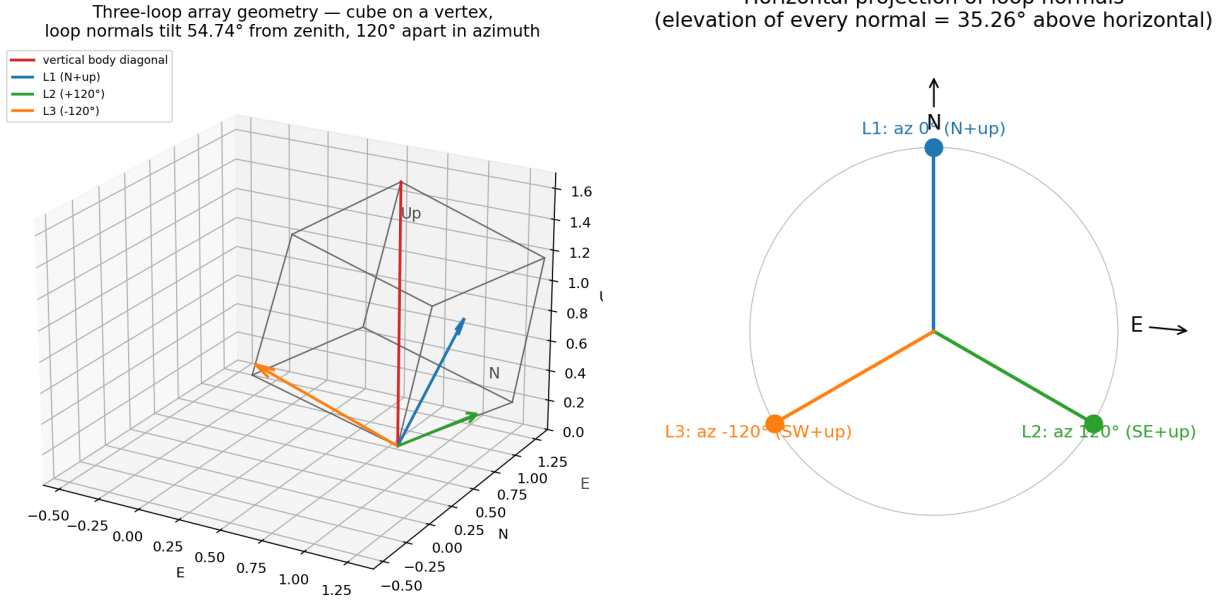
$$\begin{pmatrix} B_1(t) \\ B_2(t) \\ B_3(t) \end{pmatrix} = \mathbf{N} \mathbf{B}_{\text{lab}}(t), \quad \mathbf{B}_{\text{lab}}(t) = \mathbf{N}^{-1} \begin{pmatrix} B_1(t) \\ B_2(t) \\ B_3(t) \end{pmatrix}. \quad (3)$$

The three rows of $\mathbf{N}$ have unit length but are not mutually orthogonal — their pairwise inner product is $1/3$ (an artefact of the cube geometry). Hence one must take the proper inverse, not the transpose. $\mathbf{N}$ depends only on the array geometry, not on frequency; per-loop frequency response is a separate scalar calibration that should be folded into the B_i 's before (3) is applied.

2 Direction-dependent projection

A plane electromagnetic wave has $\mathbf{B} \perp \hat{\mathbf{k}}$ at all times, so to obtain a polarization-independent estimator of intensity arriving from direction $\hat{\mathbf{k}}$ we project the lab-frame $\mathbf{B}$ onto the plane perpendicular to $\hat{\mathbf{k}}$:

$$\mathbf{B}_\perp(t) = (\mathbf{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^\top) \mathbf{B}_{\text{lab}}(t) = \mathbf{P}(\hat{\mathbf{k}}) \mathbf{N}^{-1} \begin{pmatrix} B_1(t) \\ B_2(t) \\ B_3(t) \end{pmatrix}. \quad (4)$$



(a) 3-D view: cube on a vertex with the body diagonal vertical (red). The blue/green/orange arrows are the three loop normals $\hat{\mathbf{n}}_1, \hat{\mathbf{n}}_2, \hat{\mathbf{n}}_3$.

(b) Top-down: the horizontal projections of the three normals are 120° apart. All three normals tilt 35.26° above the horizontal.

Figure 2: Geometry of the cube-vertex three-loop array.

The polarization-independent intensity is then

$$P(\varphi, \theta; t) = |\mathbf{B}_\perp(t)|^2 = (B_1 \ B_2 \ B_3) \mathbf{M}^\top \mathbf{M} \begin{pmatrix} B_1 \\ B_2 \\ B_3 \end{pmatrix}, \quad \text{with} \quad \mathbf{M} \equiv \mathbf{P}(\hat{\mathbf{k}}) \mathbf{N}^{-1}. \quad (5)$$

Time-averaging $|\mathbf{B}_\perp|^2$ recovers a beamformed power for the chosen $\hat{\mathbf{k}}$. The instantaneous form preserves the amplitude trace of the polarization-independent envelope.

Sign flip across a loop plane. When the source elevation passes through the plane containing one of the loops, the projection coefficient $\hat{\mathbf{n}}_i \cdot \hat{\mathbf{k}}$ goes through zero and reverses sign. Because that coefficient enters the projector multiplicatively, the math handles the sign change naturally, but the corresponding loop's contribution to the *phase* of $\mathbf{B}_\perp$ flips by 180° at the crossing. Phase tracking through such transitions therefore requires either selecting a $\hat{\mathbf{k}}$ that avoids them or carrying the sign explicitly. The intensity estimator is unaffected because it is quadratic.

3 Polarimetry: the Stokes parameters

Construct two unit vectors $\hat{\mathbf{p}}, \hat{\mathbf{q}}$ that span the plane perpendicular to $\hat{\mathbf{k}}$ (Figure 3). A natural choice is

$$\hat{\mathbf{p}} = \frac{\hat{\mathbf{k}} \times \hat{\mathbf{z}}_{\text{Up}}}{|\hat{\mathbf{k}} \times \hat{\mathbf{z}}_{\text{Up}}|}, \quad \hat{\mathbf{q}} = \hat{\mathbf{k}} \times \hat{\mathbf{p}}, \quad (6)$$

so that $\hat{\mathbf{p}}$ lies in the local horizontal plane and $\hat{\mathbf{q}}$ is generally upward. After narrow-band complex baseband extraction (Section 4) we obtain three complex time-series $z_i(t)$ for the loops, which we map to the lab frame $\mathbf{Z}_{\text{lab}} = \mathbf{N}^{-1} \mathbf{z}$, project onto the plane perpendicular to $\hat{\mathbf{k}}$, $\mathbf{Z}_{\perp} = \mathbf{P} \mathbf{Z}_{\text{lab}}$, and decompose into the two complex polarization components

$$A_p(t) = \hat{\mathbf{p}} \cdot \mathbf{Z}_{\perp}(t), \quad A_q(t) = \hat{\mathbf{q}} \cdot \mathbf{Z}_{\perp}(t). \quad (7)$$

Polarization basis ($\hat{\mathbf{p}}$, $\hat{\mathbf{q}}$) for arrival at $\text{az}=273.0^\circ$, $\text{el}=12.0^\circ$

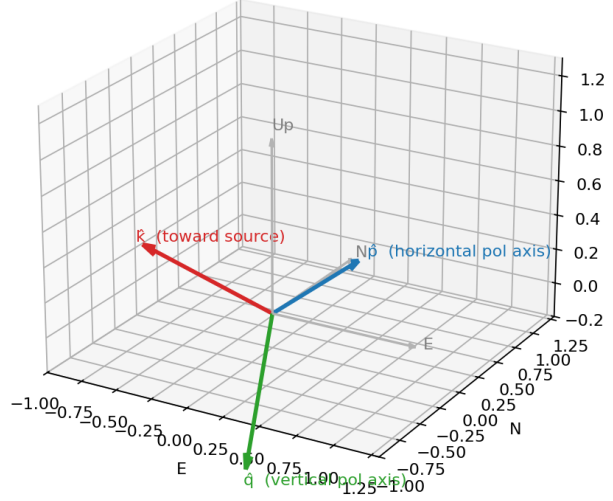


Figure 3: Polarization basis at the example arrival direction $\varphi = 273^\circ$, $\theta = 12^\circ$. $\hat{\mathbf{p}}$ lies in the local horizontal plane; $\hat{\mathbf{q}}$ is the orthogonal direction in the wave plane.

The Stokes parameters are the standard quadratic combinations:

$$\begin{aligned} I &= |A_p|^2 + |A_q|^2, \\ Q &= |A_p|^2 - |A_q|^2, \\ U &= 2 \Re[A_p^* A_q], \\ V &= 2 \Im[A_p^* A_q]. \end{aligned} \quad (8)$$

The polarization fraction is $\Pi = \sqrt{Q^2 + U^2 + V^2}/I$, the position angle of the linear part is $\chi = \frac{1}{2} \arctan(U, Q)$, and the ellipticity is $\eta = \frac{1}{2} \arctan(V/\sqrt{Q^2 + U^2})$ (linear if $\eta = 0$, RCP if $\eta = +45^\circ$, LCP if $\eta = -45^\circ$).

4 Implementation: pipeline and code

The accompanying Python module `three_loop.py` implements the pipeline:

1. Read (t, B_1, B_2, B_3) for the three loops.
2. For each loop, find the spectral peak nearest a user-supplied target carrier f_0 within a search window $\pm BW/2$.

3. Mix to baseband at the detected peak frequency and apply a brick-wall low-pass filter at $\pm BW/2$ to obtain narrow-band complex signals $z_i(t)$.
4. Form $\mathbf{Z}_{\text{lab}} = \mathbf{N}^{-1} \mathbf{z}$, $\mathbf{Z}_{\perp} = \mathbf{P}(\hat{\mathbf{k}}) \mathbf{Z}_{\text{lab}}$.
5. Compute the polarization-independent intensity $|\mathbf{Z}_{\perp}(t)|^2$.
6. Decompose onto $(\hat{\mathbf{p}}, \hat{\mathbf{q}})$, compute Stokes, polarization fraction, ellipticity, position angle.
7. Estimate the dominant polarization eigenvector and return the instantaneous amplitude, phase, and frequency on that channel.

The signature is

```
from three_loop import analyze
out = analyze(t, B1, B2, B3,
              f0,          # target carrier (Hz)
              BW,          # search/extract bandwidth (Hz)
              az_deg,      # initial guess azimuth, deg from N CW
              el_deg)      # initial guess elevation, deg above horizon
# out: amp_dominant, instant_phase, instant_freq, intensity,
#      stokes (I,Q,U,V), pol_fraction, ellipticity_deg,
#      position_angle_deg, ...
```

5 Validation: simulated WWV at 20 MHz from Fort Collins

The simulator `simulate_wwv()` synthesizes a known plane wave arriving from a chosen direction with a chosen polarization, optionally including a constant or rotating Faraday angle and additive Gaussian noise. We use it to validate the analysis pipeline.

The great-circle bearing from Fort Collins, CO (40.59° N, 105.08° W) to Cambridge, MA (42.37° N, 71.11° W) is 273° (slightly N of due west). For a single-hop F-region sky-wave reflecting at ~ 300 km altitude over a ~ 2840 km great-circle distance, the takeoff/arrival elevation is $\theta = \arctan(300/(2840/2)) \approx 12^\circ$. We therefore inject a wave with $\varphi = 273^\circ$, $\theta = 12^\circ$, an IF representation of the 20 MHz carrier, and various polarization states.

6 Practical caveats

- The geometry matrix $\mathbf{N}$ must be calibrated against the actual mechanical orientation of the loops — a few-degree tilt error causes few-percent intensity miscalibrations and few-degree position-angle biases.
- Each loop’s effective area, gain, and phase response at the operating frequency must be matched. In practice this is the dominant calibration burden; a known-direction CW beacon (or a stable HF transmission of known polarization) can be used to solve for per-channel complex gain corrections.
- The single-point three-loop array has a $P(\hat{\mathbf{k}}) = P(-\hat{\mathbf{k}})$ ambiguity: front and back are indistinguishable. Resolving this requires either an additional E-field probe or a phased pair of three-loop triplets.
- For unknown arrival direction, beamforming over a grid of (φ, θ) is suboptimal; the cleanest direction estimator is the eigendecomposition of the 3×3 covariance $\langle z_i z_j^* \rangle$, whose smallest-eigenvalue eigenvector is $\pm \hat{\mathbf{k}}$ (modulo the front/back ambiguity above).

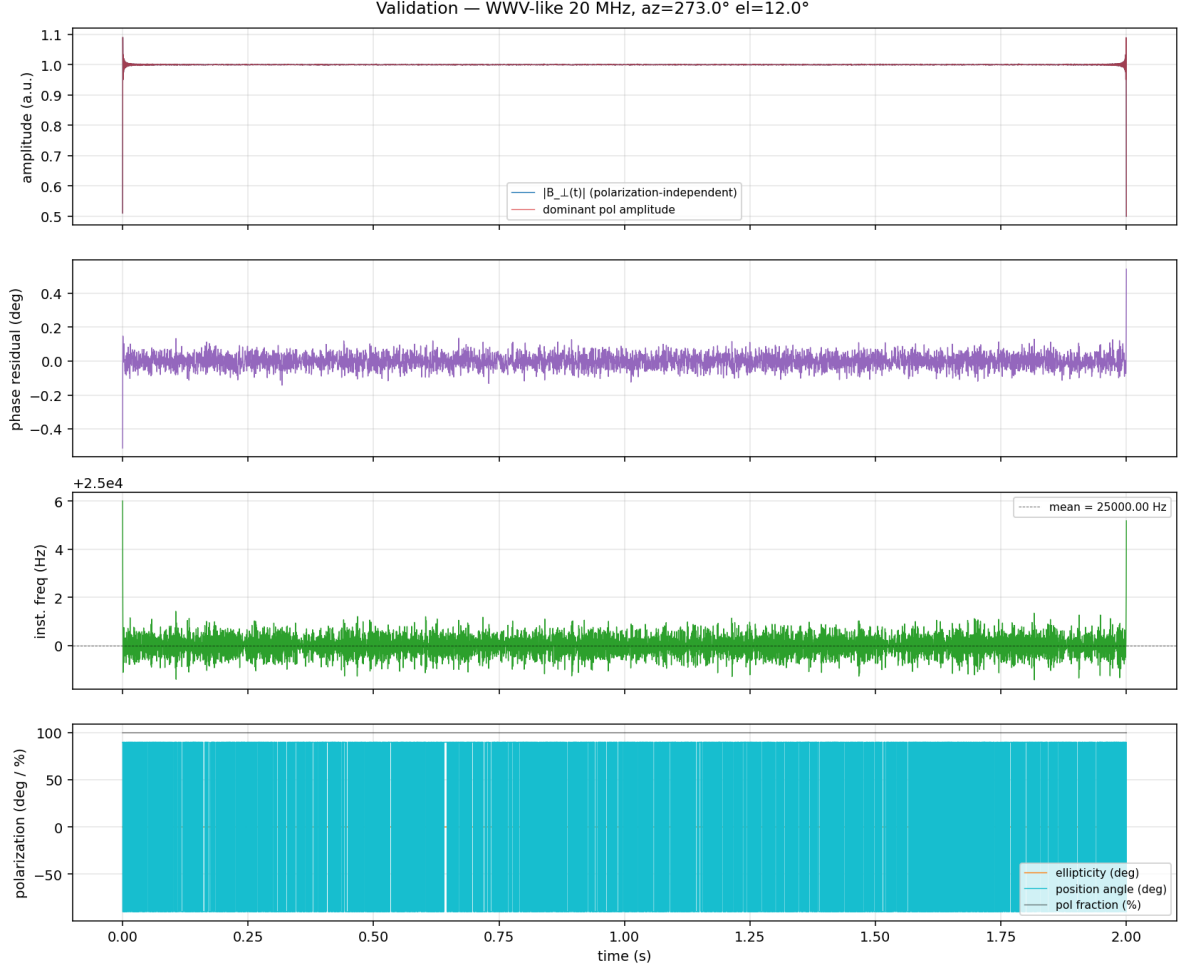


Figure 4: Validation run: linear-vertical polarization injected at $\varphi = 273^\circ$, $\theta = 12^\circ$, recovered with no Faraday rotation, SNR = 40 dB per channel. The recovered carrier frequency matches the truth to better than a millihertz over the 2-second recording; the polarization fraction is 1.000, the ellipticity is 0° (linear, as injected), and the position angle is -89.9° relative to $\hat{\mathbf{p}}$, i.e. along $-\hat{\mathbf{q}}$ which is “vertical” in the perpendicular plane.

- All three loops must be sampled *coherently* (shared LO and ADC clock). An incoherent recording set — e.g. three independent KiwiSDRs — does not in general preserve the relative phase information needed for polarimetry.

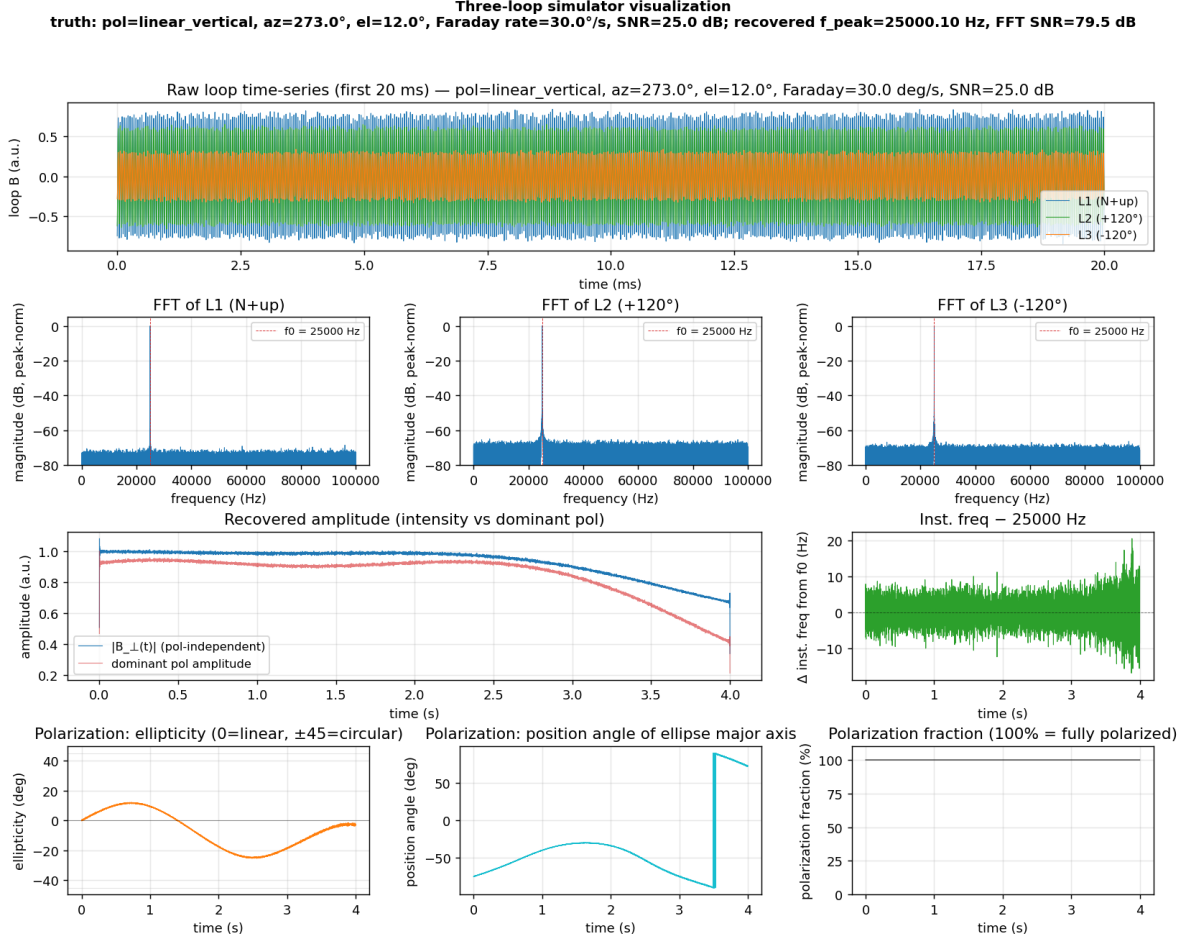


Figure 5: Linear-vertical input with Faraday rotation at $30^\circ/\text{s}$ and per-channel SNR of 25 dB. Top: raw loop time-series (first 20 ms). Second row: per-loop FFT spectra; the ~ 25 kHz IF tone is clearly visible above the noise floor in all three loops. Third row: recovered amplitude (the polarization-independent intensity is constant to within noise, while the dominant-pol amplitude shows the characteristic Faraday $|\cos \Omega(t)|$ modulation). Fourth row: polarization observables — ellipticity stays near 0° (linear throughout), the position angle sweeps $30^\circ/\text{s}$ as designed, and the polarization fraction stays near 100 %.

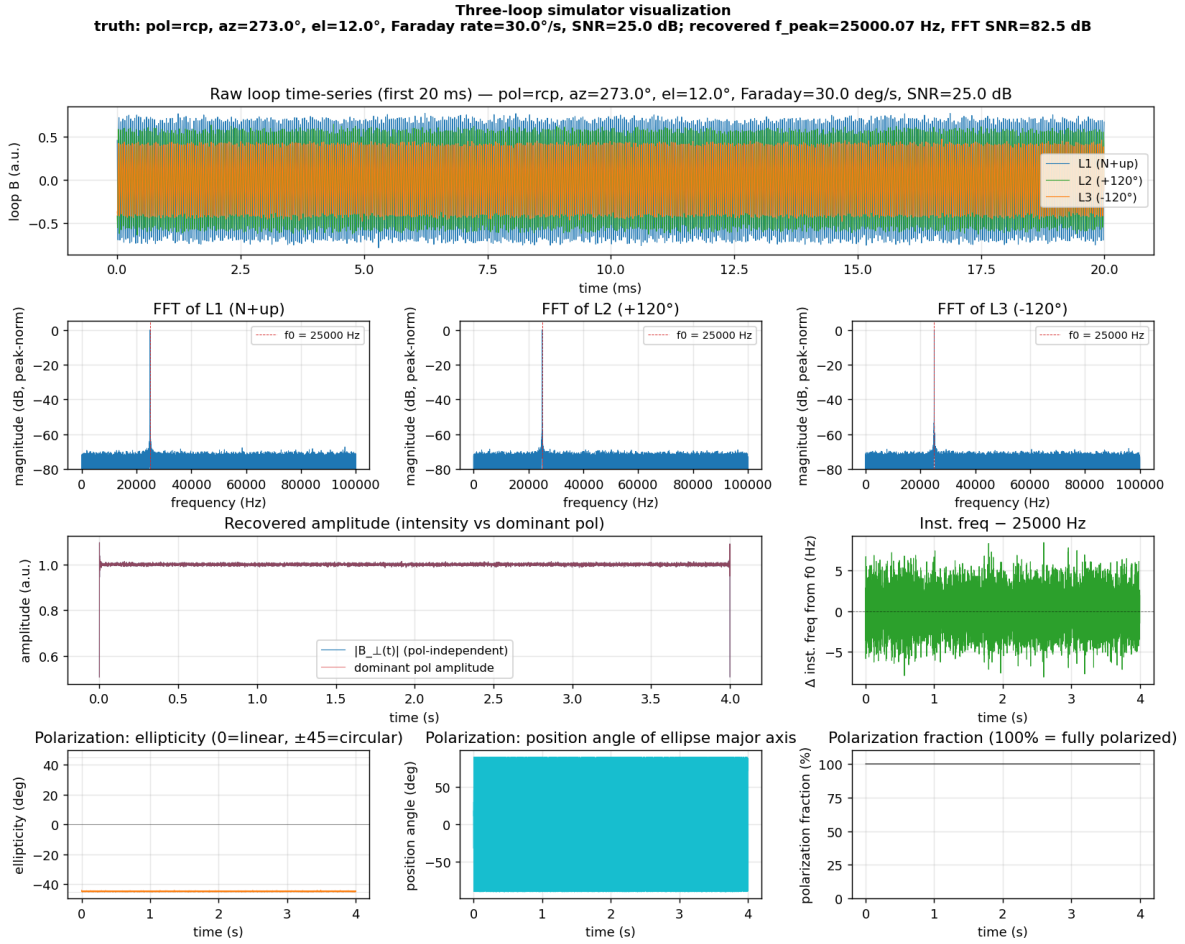


Figure 6: RCP input with the same 30°/s Faraday rotation rate. Now the polarization-independent intensity *and* the dominant-pol amplitude are constant (Faraday rotation just adds a common phase to a circularly polarized wave), the ellipticity sits at $\pm 45^\circ$, and the position-angle estimator becomes ill-defined as expected for a circular wave.